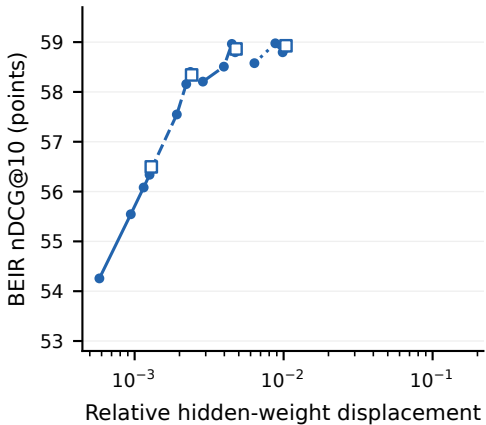


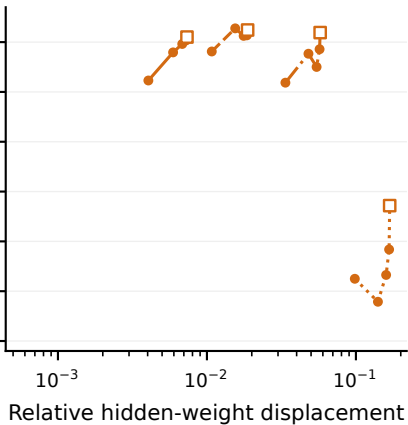
AdamW



Learning rate

- $1e-06$ (solid blue line with circles)
- $3e-06$ (dashed blue line with circles)
- $1e-05$ (solid blue line with squares)
- $3e-05$ (dotted blue line with circles)

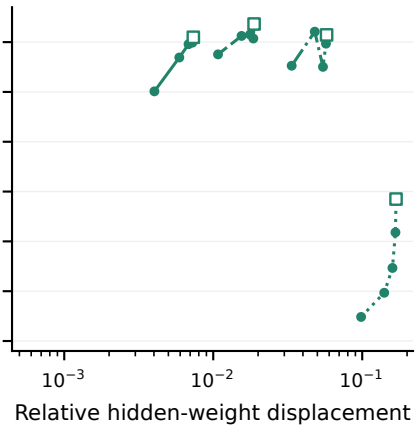
Muon



Learning rate

- $1e-04$ (solid orange line with circles)
- $3e-04$ (dashed orange line with circles)
- $1e-03$ (solid orange line with squares)
- $3e-03$ (dotted orange line with circles)

NorMuon



Learning rate

- $1e-04$ (solid green line with circles)
- $3e-04$ (dashed green line with circles)
- $1e-03$ (solid green line with squares)
- $3e-03$ (dotted green line with circles)